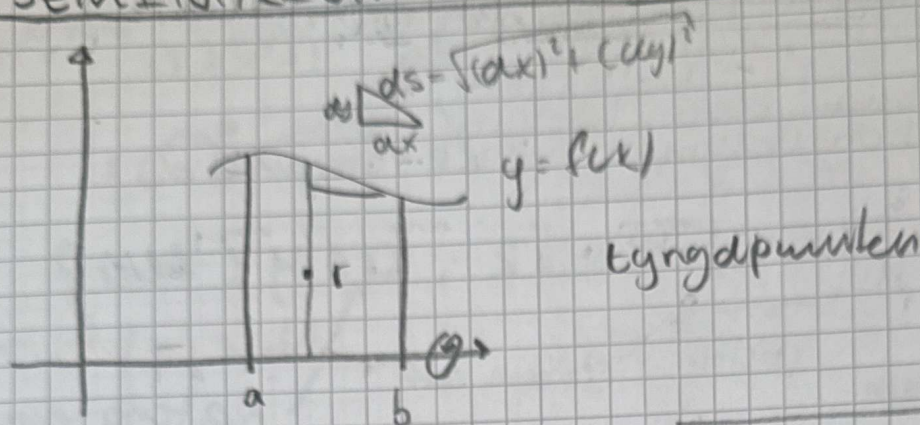


ENVARIABELS ANALYS DEL 2

SEMINARIUM 13: ROTATIONSAREA



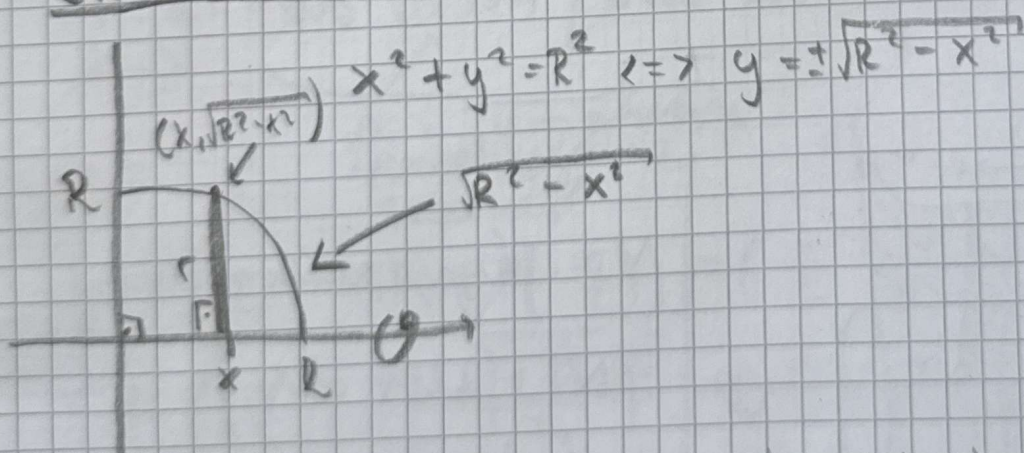
- bågens längd $ds = \sqrt{dx^2 + dy^2}$
- Tyngdpunkten avstånd till rotationsaxeln betecknas med r
- Tyngdpunktens vägg: $2\pi r$

PAPPOS - GOULDINS REGEL för rotationsyta

$$A = \int (\text{tyngdpunktens vägg}) \cdot ds =$$

$$= \int_a^b 2\pi r \, ds = \int_a^b 2\pi r \sqrt{dx^2 + dy^2}$$

Ex: Stärens Area



Lösning: $A = \int (\text{tyngdpunktens vägg}) \cdot ds$

- $ds = \sqrt{dx^2 + dy^2}$

- Tyngdpunktens vägg $= 2\pi r = 2\pi \sqrt{R^2 - x^2}$

$$A = 2 \int_0^R 2\pi \sqrt{R^2 - x^2} \cdot \sqrt{dx^2 + dy^2}$$



$$\left[\begin{aligned} &= \sqrt{(dx)^2 + (dy)^2} = \sqrt{\frac{(dx)^2}{\left(\frac{dx}{dx}\right)^2} + \frac{(dy)^2}{\left(\frac{dy}{dx}\right)^2}} \sqrt{(dx)^2} \\ &= \sqrt{1 + (f'(x))^2} dx \\ &= \sqrt{(dx)^2} = |dx| = dx, \quad x \geq 0 \end{aligned} \right]$$

$$\begin{aligned} &= 2 \int_0^R 2\pi \sqrt{R^2 - x^2} \cdot \sqrt{1 + (f'(x))^2} dx \\ &= \dots = 4\pi \int_0^R R dx = 4\pi R [x]_0^R \end{aligned}$$

$$= 4\pi R (R - 0) = 4\pi R^2$$

$$\underline{\text{SVAR: } 4\pi R^2 \text{ a.c.}}$$